

Masters Abroad on a Research Portfolio

A complete, honest plan: from QuantAura + one paper to a funded Masters seat

Prepared for: Abdullah Khan | **Profile:** BS Computer Science, 7 semesters complete, CGPA 2.5 (projected final 2.6-2.7 / 4.0) | **Assets:** QuantAura (deployed 4-model ML trading-intelligence platform, quanta.tech), one research paper at submission-readiness with 10 result tables, 20 registry-verified references, out-of-sample + cross-exchange validation, and a public reproducible repository.

How to read this document. Section 0 is the honest assessment - read it first and accept it, because every later recommendation follows from it. Sections 1-2 are what to do in the next 90 days. Section 3 is where to apply, sorted by realism, with the closed doors named explicitly so no money is wasted on them. Sections 4-8 are the operational details: documents, tests, money, timelines, interviews, integrity rules. The final page is a single master checklist.

Every scholarship's rules change yearly. Where this plan states a cutoff or deadline it reflects typical recent cycles; the action item is always: **verify on the official page before spending anything.**

0. The honest position assessment

Your asset is rare; your constraint is real. The plan is built on both.

What you have (genuinely uncommon at BS level)

- A **submission-ready empirical finance paper** whose headline result survives out-of-sample testing, 5 walk-forward folds, 15 parameter specifications, yearly subsamples, non-overlapping-event re-testing, and replication on a second exchange in a second quote currency - plus an honest negative result (statistical predictability without profitability) that reads as senior-researcher judgment.
- A **live deployed system** (quanta.tech): 4-architecture ensemble incl. a KAN, continuous learning behind a quality gate, CI/CD, and a public 'Signal Reality Check' that publishes the model's limits - a demonstrated ethic, not just a claim.
- A **public repository** where every number in the paper is reproducible by command.

What the CGPA does (say it plainly)

2.5-2.7/4.0 converts to roughly 62-67% (second division). Many programs screen transcripts **before any human reads your paper**. A hard filter cannot be argued with; the strategy is to route around filters, not to fight them.

- **Effectively closed regardless of the paper:** funded seats at US top-30 programs, Oxford/Cambridge/Imperial funded routes, ETH/EPFL, Commonwealth Shared (needs upper second), Korea GKS (~80% / top 20% cutoff), DAAD EPOS (grade-heavy), and most 'merit scholarship' schemes with published percentage cutoffs. Applying to these burns money and morale. Skip them this cycle.
- **Open:** routes where a human with discretion decides - professor-funded research seats, publication-weighted national schemes (notably China CSC), holistic/partner-country schemes (Hungary, Türkiye - borderline, Taiwan, Malaysia, Brunei, Romania via HEC), **income-based** funding (Italy's DSU regional grants ignore grades), low-tuition admission + work routes (Germany public unis + HiWi research jobs), and the two-step route (work/research year -> second paper -> reapply upgraded).

The three principles every action below follows

- **1. Portfolio-first targeting.** Only spend effort where a paper + live system can actually outweigh a transcript: research-track admissions, professors with grant money, publication-weighted schemes, income-based schemes.
- **2. Champion-driven admissions.** A professor who wants you can override or argue almost any soft cutoff. Cold outreach with the SSRN link is therefore a first-class application channel, not a nice-to-have.
- **3. Two-intake optionality.** Run Fall 2027 applications seriously, but build the Fall 2028 profile in parallel (job/RA + second paper + high GRE). If 2027 lands - great. If not, 2028-you is a different candidate.

1. Next 90 days (Aug - Oct 2026) - the foundation sprint

Academic (highest leverage per hour)

- **8th semester GPA:** treat it as a job. Every 0.1 matters twice: it lifts the final CGPA and it creates the 'strong upward trajectory' sentence your SOP needs. Target the maximum your remaining credits allow.
- **Transcript math:** compute now exactly what final CGPA each 8th-sem outcome produces. Know your number before writing any SOP.
- **FYP defence:** rehearse the live demo (site -> Reality Check -> research page); pre-answer the viva notes in research/thesis_chapter.md. An 'A' on the FYP is transcript evidence that the portfolio is yours.

The paper (your single biggest lever)

- Week 1-2: supervisor meeting - co-authorship, order, university publication rules. Bring PUBLICATION_README.md as the agenda.
- Week 2-3: assemble the manuscript from thesis_chapter.md + model_findings.md; do YOUR verification pass - open all 20 DOI links, re-run the four scripts, be able to derive every table.
- Week 3-4: **SSRN preprint** (free, no endorsement needed, citable link in ~1 week). This link goes into every application and email below - it is the difference between claiming and showing.
- Week 4-6: submit to Finance Research Letters (no fee). Decision timelines run months - the SSRN link does the application-season work meanwhile. 'Under review at FRL' is a legitimate CV line.

Owning the material (non-negotiable)

- One full week: re-run every script, re-derive the matched control on paper, explain the Goodhart finding aloud, answer all viva-note questions without looking. Every interview below will test whether the work is yours.

Tests - book NOW (seats fill)

Test	Target	Why this number	Cost (approx.)	When
IELTS Academic	7.0+ overall, no band under 6.5	Clears every scheme in Section 3; 7.5 helps Europe	~PKR 65-70k	Book Aug, sit Sep-Oct
GRE General	315+ (Q 165+) - OPTIONAL this cycle	Only some routes need it; a high Q partially offsets GPA. Skip if cash-tight; mandatory for the 2028 US route.	US\$220	Nov-Dec if taken

Digital presence (admissions committees do google you)

- GitHub: pin the FYP repo; README with the paper abstract, architecture diagram, reproduction commands, screenshots.
- LinkedIn: headline 'BS CS | Author: Bitcoin regime-propagation research (SSRN) | Built quantaura.tech'; About section = paper pitch paragraph.
- Record a 3-minute screen-capture demo video (site tour -> one finding -> repo). Unlisted YouTube; link in CV and emails.
- Keep quantaura.tech alive all season - a dead demo link in March kills the story. Renew domain/server now.

2. The application arsenal (build once, reuse everywhere)

CV (2 pages max) - exact order for your profile

- 1. Header + links (site, SSRN, GitHub, video). 2. **Research** (paper first: title, venue status, one-line finding). 3. **Projects** (QuantAura: 5 bullets - system, honest-metrics feature, continuous-learning gate, research page, deployment). 4. Skills. 5. Education (degree, university, expected date - **CGPA only if the application requires it**; never volunteer it in the CV). 6. Awards/activities if any.
- Bullet formula: verb + artifact + number. 'Validated a cross-asset regime effect out-of-sample (+15pp over matched control, $p < 0.001$) and replicated it on a second exchange' beats any adjective.

Statement of Purpose - the paragraph map (600-800 words)

- P1 - open with the finding, not biography: the co-movement-without-leadership result in two sentences, then 'I built the system and wrote the paper that established this.'
- P2 - the research story: target redefinition, the F1-to-52% decomposition, the Goodhart discovery. Show judgment, not just effort.
- P3 - the system story: deployed, continuously learning, honest by construction. One sentence on AI-assisted workflow used transparently and verified personally - modern committees respect candor and it pre-empts suspicion.
- P4 - **the GPA paragraph (do not skip; do not grovel)**: three sentences maximum. Acknowledge, contextualize with trajectory (8th-sem grades, FYP grade), then redirect: 'the record that better predicts my research ability is the paper and the system, which are public and reproducible.' Never blame, never over-explain.
- P5 - why THIS program/professor: name 2-3 specific papers/groups and connect to concrete next questions (multi-era regime propagation, DY identification bounds, Goodhart effects in continuously-trained financial ML).
- P6 - trajectory: Masters -> PhD/quant research; one sentence on returning value.

Recommendation letters (3)

- FYP supervisor (research letter - the crucial one): brief them with a 1-page summary of results + your specific contributions; ask them to address research ability and independence explicitly, and - if true - to note the paper exceeds typical Masters theses they have supervised.
- Second: any instructor from a course you scored well in (counters the transcript narrative). Third: head of department or industry mentor. Give every writer: CV, SOP draft, paper abstract, deadline table, and a polite 2-week-before reminder.

The cold email to professors (your highest-ROI channel)

Send 25-40 of these, personalized, Sep-Nov 2026. Targets: authors citing or cited by your bibliography (crypto connectedness, financial ML, market microstructure), plus any group whose page says 'openings'. Template:

Subject: Prospective MS student - out-of-sample evidence on Bitcoin regime propagation (SSRN)

Dear Professor [Name], - I am a final-year CS student. In [their paper, year] you showed [one-sentence real engagement]. My own work may interest you: across 2024-26 data on two exchanges, Bitcoin shows almost no hourly predictive lead over altcoins ($r \sim 0.006$) yet a robust regime-level lead (+15pp out-of-sample, $p < 0.001$) that is nonetheless unprofitable after costs - SSRN: [link]. I built and deployed the full system behind it (quantaura.tech, repo attached). I am applying for [term] and would be glad to work on [one concrete extension tied to THEIR agenda]. Would you have funded openings, or advice on your program's fit? - [Name, CV link]

- Rules: under 180 words; one real sentence about THEIR work (never generic); never mention CGPA in the first email; follow up once after 10-14 days; track everything in a spreadsheet (name, uni, date, response).

Fraud warning

Never pay an 'agent' who guarantees scholarships; never pay any fee to a 'scholarship' that emails you first; official schemes never charge application fees beyond the university's own. All-caps guarantee = scam.

3. Where to apply - tiered by realism at CGPA 2.6-2.7

Ratings: **A** = genuinely winnable with your portfolio; **B** = winnable if a professor champions you or the year's competition is soft; **C** = borderline, verify cutoff before spending; **X** = skip this cycle.

Tier A - primary targets (bulk of your effort)

Scheme / Route	What it covers	Why it fits YOU	Apply via / when (typical)	Fit
China CSC (Chinese Government Scholarship) - Type A via embassy AND Type B direct to universities (e.g. Tsinghua, PKU, Fudan, USTC, Wuhan, HIT)	Full tuition + stipend + housing	Publication-weighted review at many hosts; supervisor acceptance letters carry huge weight - your cold-email channel feeds directly into this. Many strong CS/FinTech groups.	campuschina.org + university portals; HEC also announces the Pakistan allocation. Windows mostly Dec-Apr for Sep intake.	A
Hungary Stipendium Hungaricum (via HEC Pakistan nomination)	Tuition + stipend + housing + insurance	Partner-country quota, holistic-ish review, CS/finance programs (BME, ELTE, Corvinus, Debrecen). Pakistan quota is real and HEC-administered.	stipendiumhungaricum.hu + hec.gov.pk announcements; usually opens Nov, closes mid-Jan.	A
Italy - admission + DSU regional grant (EDISU/DSU Toscana/LazioDiSCo etc.) + fee reduction by family income	Regional grant ~ EUR 5-7k/yr + free meals/housing support + near-zero tuition at low income	Income-based, not merit-based - your CGPA is almost irrelevant. Many programs (Padua, Bologna, Milan-Bicocca, Trento; look for Data Science / Quantitative Finance) admit on curriculum fit + interview.	University pre-admission (University) Feb-Jun; DSU applications ~Jun-Sep. Needs income documentation (attested).	A
Professor-funded research seats in China / Malaysia (USM, UM) / Thailand (AIT fellowships)	Varies: RA stipend + tuition waivers	Decided by the person who read your paper, not by a registrar.	Direct email (Section 2 template) -> supervisor acceptance -> university formality.	A

Tier B - champion-dependent

Route	Reality check	Action	Fit
Hong Kong MPhil (PolyU, CityU, HKBU) / research assistant first	Formally 'second class honours'; a professor with GRF funding can push a strong case. MPhil is 2yr funded (~HK\$18k/mo).	Cold-email FinTech/ML-finance groups with SSRN link; ask about RA-then-MPhil route.	B
Taiwan - MOE Taiwan Scholarship + university scholarships (NTU, NTHU, NCKU)	University-set minima vary; several programs review holistically; ICDF covers development-partner tracks.	Check each university's international office page; embassy MOE window ~Feb-Mar.	B
Erasmus Mundus Joint Masters (pick 2-3 consortia max)	Each consortium sets rules; some weigh motivation+research heavily, many implicitly filter grades. Application is heavy.	Read 2-3 relevant consortia (data science / computational finance flavored) selection criteria closely; apply only where no explicit GPA floor. Window Oct-Jan.	B
Germany public universities (admission WITHOUT scholarship) + HiWi research jobs + no/low tuition	uni-assist grade conversion is harsh on a 2.6; SOME programs have no strict cutoff. Living costs need blocked account (~EUR 11-12k) - this route needs family funding for year 1.	Filter DAAD course database for programs listing no minimum; treat as self-funded-entry, research-funded-later.	B
Brunei (UBD/UTB Graduate Scholarship), Romania/Hungary bilateral seats via HEC, Malaysia MTCP	Smaller, quieter schemes; competition lower; criteria flexible-ish.	Watch hec.gov.pk 'scholarships' page monthly; apply to everything eligible.	B

Tier C - verify before spending

- **Turkiye Burslari:** postgraduate minimum has typically been ~75% (~3.0/4.0) - likely just above you. Verify the current cycle's threshold (turkiyeburslari.gov.tr, window ~Jan-Feb); apply only if you clear it on their conversion.
- **UK self-funded + partial scholarships** (e.g., Global Wales, university bursaries at post-92 or Scottish unis): admission achievable, funding partial at best. Only if family co-funding exists.

Tier X - do not spend money this cycle

- US top-30 funded MS/PhD - GPA filter + funding scarcity. Revisit in the 2028 two-step with GRE 320+, two papers, work experience.
- Oxbridge/Imperial/ETH/EPFL funded; Commonwealth Shared/Split-site; Korea GKS; DAAD EPOS; Japan MEXT university-track (embassy-track screens grades hard too); Fulbright (competitive Pakistani pool is 3.5+); Sweden SI; Chevening (needs 2+ yrs work anyway).

The two-step route (run in parallel - it is not a consolation prize)

- Aug 2026-mid 2027: land a quant-dev / ML-engineer / research-assistant job (Karachi/Lahore/Islamabad fintechs, remote firms, or university RA). Your interview stories are already better than most seniors': the Goodhart bug, the untrained-weights incident, predictability-vs-profitability.
- 2027: second paper (extensions listed in the repo README), GRE 320+, savings.
- Oct 2027-Jan 2028: reapply everywhere - now as 'published researcher with industry experience whose last academic year showed an upward trend'. This candidate wins seats the 2026 candidate cannot.

4. Documents and logistics (Pakistan-specific, start early)

Item	Details	Lead time
HEC degree attestation	Required by most foreign schemes + Italian/Chinese procedures. portal.hec.gov.pk ; courier both ways.	2-6 weeks - start at graduation
Transcripts x 8-10 sealed copies	Order from registrar in one batch; some portals need scans only, embassies need sealed.	1-3 weeks
Passport	Renew if under 18 months validity - every application asks for it.	2-4 weeks
MOFA attestation + translations	Needed for Italy (Dichiarazione di Valore / CIMEA statement) and some embassy tracks.	3-6 weeks
Income certificate + bank documents	Italy DSU requires family income documentation (attested, sometimes translated). Germany needs blocked-account proof.	4-8 weeks
Medical/police certificates	China CSC physical exam form; police clearance for several schemes.	1-2 weeks each

Application budget (plan it like a project)

Cost	Approx.	Note
IELTS	PKR 65-70k	One sitting if prepared; budget a retake
GRE (optional this cycle)	US\$220	Skip if cash-tight; mandatory for 2028-US route
University application fees	US\$0-100 each	CSC-B some free; Italy ~EUR 30-50 pre-eval; Hungary free; target 8-12 applications total
Attestations + couriers + translations	PKR 30-60k	HEC + MOFA + CIMEA/DoV if Italy
Total realistic season	PKR 250-450k	Concentrate on Tier A; every Tier-X application avoided funds a Tier-A one

5. Master timeline

When	Do
Aug 2026	Supervisor meeting; manuscript assembly + your verification pass; book IELTS; start 8th sem strong; renew domain/passport check
Sep 2026	SSRN preprint live; FRL submission; CV/SOP v1; demo video; start professor cold-emails (10/wk); HEC portal account
Oct 2026	IELTS sitting; Erasmus consortia shortlist (if any qualify); Italy program shortlist; continue outreach; track responses
Nov 2026	Stipendium Hungaricum opens - prepare; CSC university shortlist from professor replies; LoR writers briefed
Dec 2026	Final exams (protect the GPA); submit Hungary; first CSC-B portals open; GRE if taking
Jan 2027	Hungary deadline; CSC embassy (HEC) + direct applications; Turkiye window if eligible; Erasmus deadlines
Feb-Apr 2027	Remaining CSC portals; Taiwan MOE; Italy pre-admissions (University); interviews - rehearse the demo
May-Jul 2027	Results season; Italy DSU grant application immediately after any Italian admission; visa processes; graduation + attestation

When	Do
Aug-Sep 2027	Fly - or execute two-step: job secured, paper #2 outlined, GRE booked
Oct 2027-Jan 2028	(If two-step) reapply everywhere incl. previously-closed tiers with upgraded profile

6. Interviews - where your profile actually wins

At CGPA-filtered stages you are weak; in any live interview you are strong, because you own something concrete. Steer every answer to the artifacts.

- 'Tell me about your research' - 90-second version: question, method (matched control), result (+15pp OOS, two exchanges), honest limit (not profitable after costs). Practice until fluent.
- 'Why is your GPA low?' - same three sentences as the SOP paragraph, delivered without flinching, then bridge: 'which is why everything I claim is public and reproducible - may I show you?'
- Expect and rehearse: Isn't this just persistence? (control differences it out) - Why believe it isn't overfit? (OOS lift exceeded in-sample; 15/15 specs; second exchange) - What was hardest? (the inverted-control story: the wrong baseline gave the opposite conclusion) - What would you research next? (have 3 concrete extensions ready).
- Always have the live site + repo open in a tab. 'May I share my screen for 60 seconds' converts interviews.

7. Integrity rules (violating any one destroys the whole plan)

- Never round the CGPA up; never omit it where asked; never claim the paper is 'published' before acceptance - 'preprint on SSRN, under review at [journal]' is the exact truthful phrase.
- Disclose AI assistance wherever asked (paper acknowledgements, any application question about tools): you used AI assistance extensively and verified everything yourself - said plainly, once, without apology.
- Never let anyone else write your SOP answers or sit your tests; scholarship committees in Pakistan's pool actively screen for this and blacklist.
- Claims about the paper stay exactly as bounded in it: 'novel per our searches', 'not profitable after costs', '2024-26, two exchanges'. Overselling to a finance professor is instant death.

8. Risks and honest odds

- **Nothing here guarantees a scholarship.** Realistic read: Tier A gives you several genuinely winnable shots this cycle (CSC via a supervisor who wants you; Hungary's quota; Italy's income-based route is the single most probability-dense option). The two-step raises 2028 odds dramatically. The failure mode to avoid is spreading thin across prestige names that filter you out before reading page one.
- Paper rejection at FRL is likely on first submission (true for everyone) - the SSRN link keeps working regardless; resubmit down the fallback list (Digital Finance, JRFM, Financial Innovation).
- Single dependency to protect: the supervisor relationship. Their letter + co-authorship + endorsement is load-bearing for CSC, HK, and every professor route. Invest in it accordingly.

Master checklist (print this page)

This week

- ☐ Meet supervisor: co-authorship + university publication rules
- ☐ Book IELTS seat
- ☐ Compute exact final-CGPA scenarios
- ☐ Pin repo + rewrite GitHub README with abstract and screenshots

This month

- ☐ Manuscript assembled; personal verification pass of all 20 DOIs and 4 scripts
- ☐ SSRN preprint submitted
- ☐ CV v1 + SOP v1 (with the GPA paragraph)
- ☐ 3-minute demo video recorded
- ☐ First 10 professor emails sent + tracking sheet

This quarter

- ☐ IELTS 7.0+ achieved
- ☐ FRL submission in
- ☐ 25-40 professors contacted, follow-ups done
- ☐ Hungary application prepared; CSC shortlist built from replies
- ☐ HEC attestation started; 8-10 sealed transcripts ordered
- ☐ Italy shortlist + income documents started
- ☐ 8th-sem midterms maximized

Application window (Dec-Apr)

- ☐ Hungary submitted (Jan)
- ☐ CSC embassy + 3-5 direct universities submitted
- ☐ Taiwan / Malaysia / Brunei / HEC-announced bilaterals as eligible
- ☐ Italy pre-admissions; DSU documents ready
- ☐ Every interview: demo rehearsed, artifacts open

Insurance (parallel)

- ☐ Job applications out by Mar 2027 (quant dev / MLE / RA)
- ☐ Paper #2 topic chosen from repo extension list
- ☐ GRE plan for the 2028 route

Final word: the plan's engine is not the CGPA and never will be - it is the paper, the live system, and your ability to defend both fluently. Everything in Sections 1-2 exists to make those three things impossible to ignore.